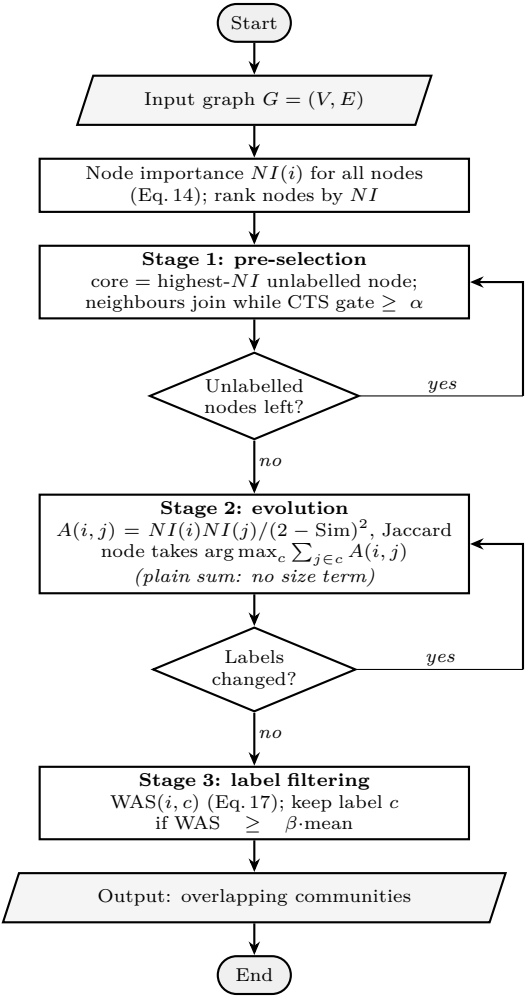
Start
Input graph $G = (V, E)$
Node importance $NI(i)$ for all nodes (Eq. 14); rank nodes by NI
Stage 1: pre-selection
core = highest-NI unlabelled node; neighbours join while CTS gate $\geq$ α
Unlabelled nodes left?
yes
no
Stage 2: evolution
$A(i,j) = NI(i)NI(j)/(2 - \mathrm{Sim})^2$, Jaccard
node takes $\arg\max_c \sum_{j \in c} A(i,j)$
(plain sum: no size term)
Labels changed?
yes
no
Stage 3: label filtering
WAS(i, c) (Eq. 17); keep label c if WAS $\geq$ β·mean
Output: overlapping communities
End